

DRM Pulsed-Flow KPI Pipeline

Quick-Start Guide for Running the Script

This guide explains how to set up your data and run the DRM_PULSING_DATA_SCRIPT notebook - no coding knowledge needed. Just follow the steps in order.

1. What you need before you start

- Your raw instrument files for each catalyst (see naming rules below).
- One folder on your computer where ALL of these files will sit together.
- Jupyter Notebook (or JupyterLab) installed, with the notebook file DRM_PULSING_DATA_SCRIPT.ipynb.

2. Name your files correctly

Every catalyst needs exactly 3 files, and all 3 must share the same ID at the start of the filename. The script uses this shared ID to automatically match them together - it does not matter what the ID is (a number, a code name, etc.), as long as all three files for one catalyst use the exact same one.

File name pattern	What it contains
<ID>_bypass.dat	Feed gas measured with NO catalyst in the reactor (the reference / baseline run).
<ID>_reaction.dat	The actual reaction run, measured over the catalyst bed.
<ID>_parameters.dat	The process log: reactor temperatures, pressure, and gas flow setpoints.

Example - two catalysts, ready to go:

Files naming
Catalyst_02_bypass.dat
Catalyst_02_reaction.dat
Catalyst_02_parameters.dat
Catalyst_05_bypass.dat
Catalyst_05_reaction.dat
Catalyst_05_parameters.dat

Important: keep the underscore and the exact words "bypass", "reaction", and "parameters" - the script looks for these exact words to know which file is which.

3. Put everything in ONE folder

Place every catalyst's files - all of them, for every catalyst you want to analyze - into a single folder. You do not need to create subfolders per catalyst; the script sorts them out automatically using the file names.

Example folder layout:

```
C:\Users\YourName\Documents\DRM_Project\Raw_Data\  
    Catalyst_02_bypass.dat
```

```
Catalyst_02_reaction.dat
Catalyst_02_parameters.dat
Catalyst_05_bypass.dat
Catalyst_05_reaction.dat
Catalyst_05_parameters.dat
```

4. Tell the script where that folder is

Open DRM_PULSING_DATA_SCRIPT_Ar.ipynb in Jupyter. Near the top of the code, you will see two lines like this:

```
INPUT_FOLDER = r"C:\Users\YourName\Documents\DRM_Project\Raw_Data"
OUTPUT_FOLDER = r"C:\Users\YourName\Documents\DRM_Project\Results"
```

1. INPUT_FOLDER - paste the path to the folder from Step 3 (where your raw files are).
2. OUTPUT_FOLDER - paste the path to any folder where you want the results saved (**it can be the same folder**, or a different one - the script will create it if it doesn't exist).

Tip: to copy a folder path on Windows, open the folder in File Explorer, click the address bar once, then copy the highlighted text.

5. Run the script

1. In Jupyter, click on the code cell.
2. Click "Run" (or press Shift + Enter).
3. Wait for it to finish - you'll see a short progress message printed for each catalyst it finds and processes.

You do not need to change anything else in the script for a normal run.

6. What you'll get back

Everything appears in your OUTPUT_FOLDER once the run finishes:

Output file	What it's for
<ID>_integrated_results.xlsx	One Excel file per catalyst: merged raw data, smoothed data, and the final calculated KPIs (conversion, yield, selectivity, temperature, etc.).
<ID>_averaging_qc_bypass.png <ID>_averaging_qc_reaction.png	Quality-check pictures per catalyst - show the raw noisy signal next to the smoothed line, so you can visually confirm the smoothing worked well before trusting the results.
Master_DRM_LookerStudio_Data.csv	One combined table with every catalyst's KPIs together - useful for Excel, Looker Studio, Power BI, etc. This is not really relevant for you, I needed it for other purposes.
DRM_Interactive_Dashboard.html	Open this in any web browser - it shows every catalyst overlaid together on interactive charts, so you can compare them directly. It also shows you the averaging results to see how well the averaging worked.

7. How to know it worked well

- Open the *_averaging_qc_reaction.png and *_averaging_qc_bypass.png pictures for each catalyst - the smoothed (bold) line should look flat and steady in the middle and end sections, not still wiggling up and down.

- Most preferably **Open DRM_Interactive_Dashboard.html** in a browser and check the catalysts look reasonable and comparable.
- Watch the messages printed while the script runs - if something looks off (for example a warning about the pulse timing), it will be printed there in plain language.

8. Common questions

“Can I run more than one catalyst at once?”

Yes - just put all catalysts' files in the same input folder. The script automatically finds and processes every complete set it finds, and compares them together in the dashboard.

“What if I'm missing one of the three files for a catalyst?”

That catalyst will simply be skipped (it needs all three: **bypass, reaction, and parameters**). Double check the file naming if a catalyst you expected doesn't show up in the results.

“Do I need to rename my raw instrument export files?”

Yes - rename them to follow the *Catalyst_<ID>_bypass.dat* / *Catalyst_<ID>_reaction.dat* / *Catalyst_<ID>_parameters.dat* pattern before running the script, using the same <ID> for all three files of one catalyst.